Assignment 1

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- When I knit my code, all the values appear differently in the knit file from the values in RStudio, so I manually wrote down answers in the knit file instead of calling variable names directly.

Problem 1

```
# Read data from csv file
library(caret)

GPA.data <- read.csv("/cloud/project/Data/Grade Point Average Data.csv")
```

a-)

```
set.seed(1234) # to get the identical result for each data partition attempt
DataSplit <- createDataPartition(y = GPA.data$Y, p = 0.7, list = FALSE)

TrainData <- GPA.data[DataSplit,] # development sample (70%)
TestData <- GPA.data[-DataSplit,] # hold-out sample (30%)
```

b-)

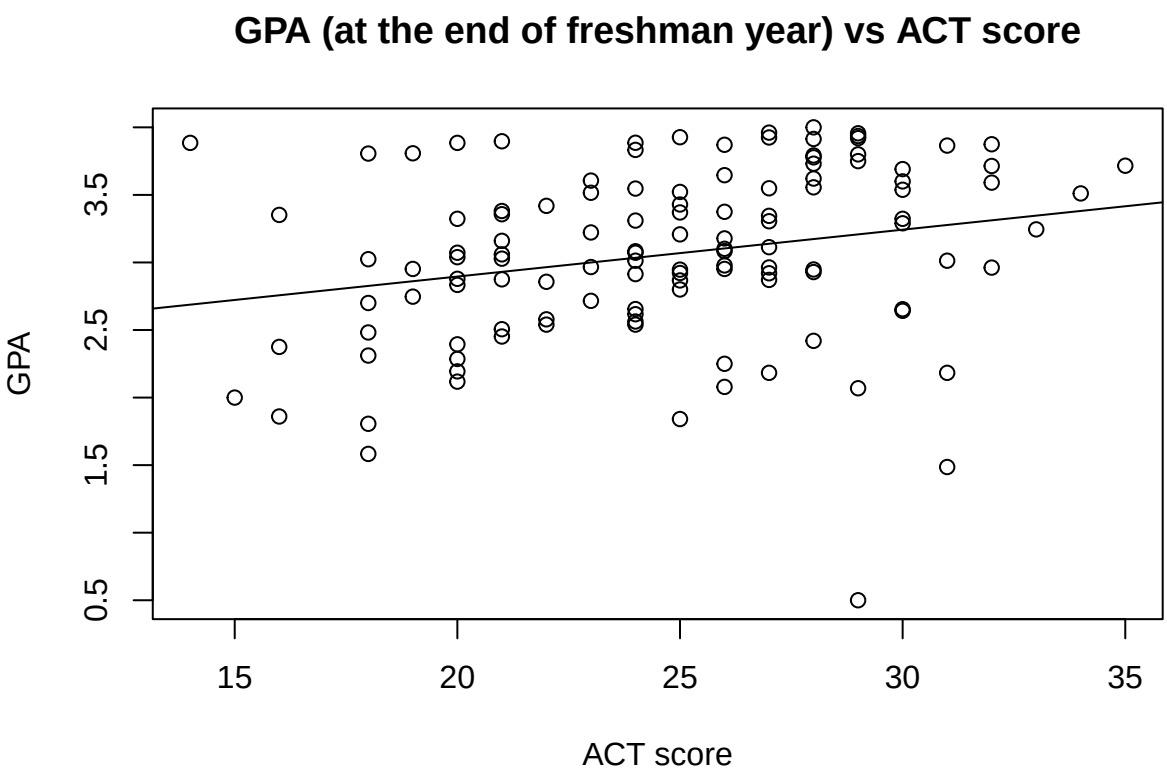
```
LmFit <- lm(Y ~ X, data = TrainData)
b_0 <- LmFit$coefficients[1]
b_1 <- LmFit$coefficients[2]
```

b_0 and b_1 represent estimates of β_0 and β_1, respectively.

$b_0 = 1.922$
$b_1 = 0.0473$
Y = 1.922 + 0.0473X

c-)

```
plot(GPA.data$X, GPA.data$Y, main = "GPA (at the end of freshman year) vs ACT score",
     xlab = "ACT score", ylab = "GPA") + abline(LmFit)
```




```
## integer(0)
```

Because many data points deviate much from the estimated regression line, it does not appear to fit the data well. The very low R-squared value, 0.1087, also supports this claim.

d-)

```
y_hat <- b_0 + b_1 * 30
```

Estimate of GPA (at X = 30) = 3.3418

e-) The point estimate of the change in the mean response when ACT score increases by one point is equal to 0.0473, which is equivalent to b_1.

f-)

```
sum(LmFit$residuals)
```

```
## [1] 1.797174e-15
```

Although the sum of residuals shown above is not exactly zero, it is negligibly small, so we can say that they sum to zero.

g-)

```
n <- nrow(TrainData) # number of data points in the development sample
var <- sum(LmFit$residuals^2)/(n-2) # summary(LmFit)$sigma^2 gives the same result.
sd <- sqrt(var)
```

Estimates of σ^2 and σ are as below:

$s^2 = 0.3356$
$s = 0.5793$

σ is expressed in GPA.

h-)

```
summary(LmFit)
```

```
##
## Call:
## lm(formula = Y ~ X, data = TrainData)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7079 -0.3458  0.1102  0.4489  0.9796
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.20222    0.41201   5.345 7.96e-07 ***
## X            0.03468    0.01628   2.130   0.0361 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6523 on 82 degrees of freedom
## Multiple R-squared:  0.05244,    Adjusted R-squared:  0.04088
## F-statistic: 4.538 on 1 and 82 DF,  p-value: 0.03615
```

```
y_hat <- predict(LmFit, TestData)

train_model <- data.frame(obs = TrainData$Y, pred = LmFit$fitted.values)
defaultSummary(train_model)
```

```
##       RMSE   Rsquared        MAE
## 0.64448309 0.05243993 0.49182580
```

```
test_model <- data.frame(obs = TestData$Y, pred = y_hat)
defaultSummary(test_model)
```

```
##      RMSE  Rsquared       MAE
## 0.5526029 0.1439106 0.4441271
```

(RMSE, Rsquared, MAE values are different in the knit file from those in RStudio. I used RStudio values as answers.) The linear regression model 'LmFit' does not appear to be a very good model as R-squared is very low (0.1087). Furthermore, when this model is tested with the hold-out sample, R-squared decreases even more to 0.0277 and RMSE increases by about 0.14. Considering the significant drop in R-squared, we should be worried about overfitting, and it seems further analysis is necessary. The p-value of X indicates that the predictor variable, ACT test score, is significant, and we can find a weak positive linear relationship between X and Y.

Problem 2

```
# Read data from csv file
CDIdata <- read.csv("/cloud/project/Data/CDI Data.csv")
```

a)

```
# Assign variables in simple names
Y <- CDIdata$Number.of.active.physicians
X1 <- CDIdata$Total.population
X2 <- CDIdata$Number.of.hospital.beds
X3 <- CDIdata$Total.personal.income
```

```r
CDI_sub <- data.frame(cbind(Y,X1,X2,X3)) # New data

set.seed(1234) # to get the identical result for each data partition attempt
train_index <- createDataPartition(y = CDI_sub$Y, p = 0.7, list = FALSE)
train_set <- CDI_sub[train_index,] # development sample (70%)
test_set <- CDI_sub[-train_index,] # hold-out sample (30%)
```

b)

```r
# Model with Total.population as predictor variable
fit_tp <- lm(Y ~ X1, data = train_set)
b0_tp <- fit_tp$coefficients[1]
b1_tp <- fit_tp$coefficients[2]
```

X_1 = Total.population
Y = -93.1566 + 0.0028X_1

```r
# Model with Number.of.hospital.beds as predictor variable
fit_hb <- lm(Y ~ X2, data = train_set)
b0_hb <- fit_hb$coefficients[1]
b1_hb <- fit_hb$coefficients[2]
```

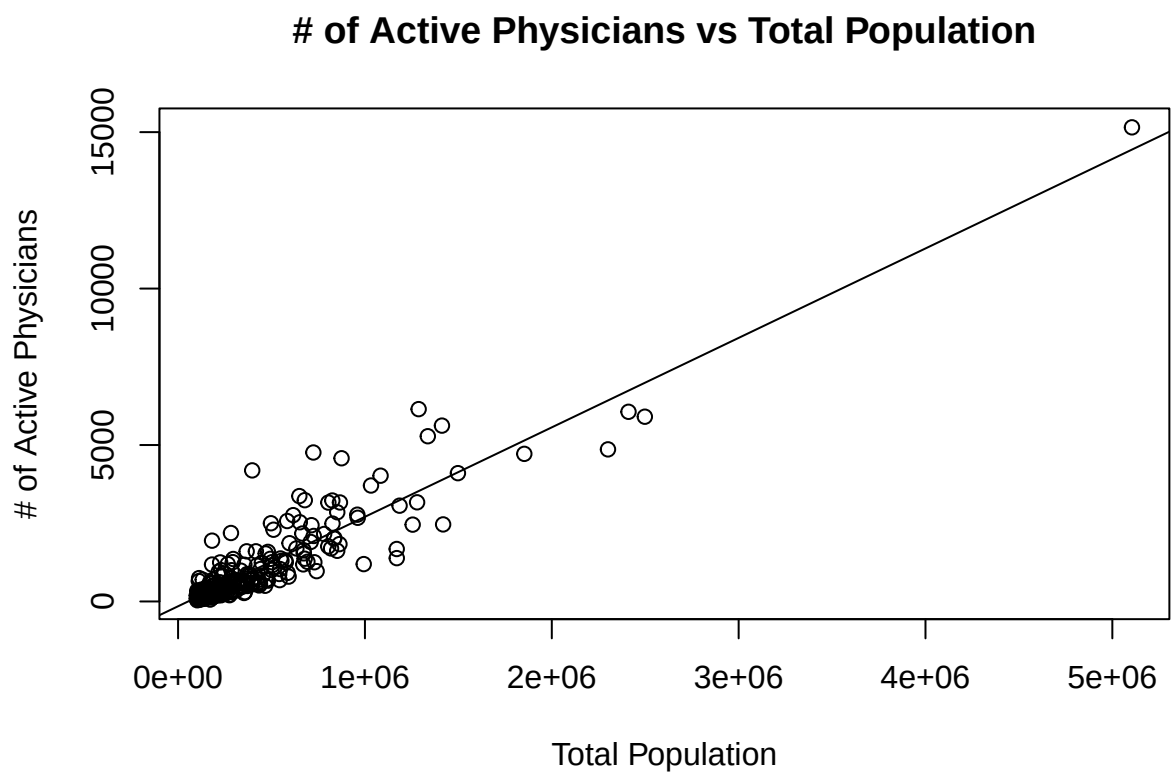
X_2 = Number.of.hospital.beds
Y = -105.0845 + 0.7436X_2

```r
# Model with Total.personal.income as predictor variable
fit_pi <- lm(Y ~ X3, data = train_set)
b0_pi <- fit_pi$coefficients[1]
b1_pi <- fit_pi$coefficients[2]
```

X_3 = Total.personal.income
Y = -40.2525 + 0.1318X_3

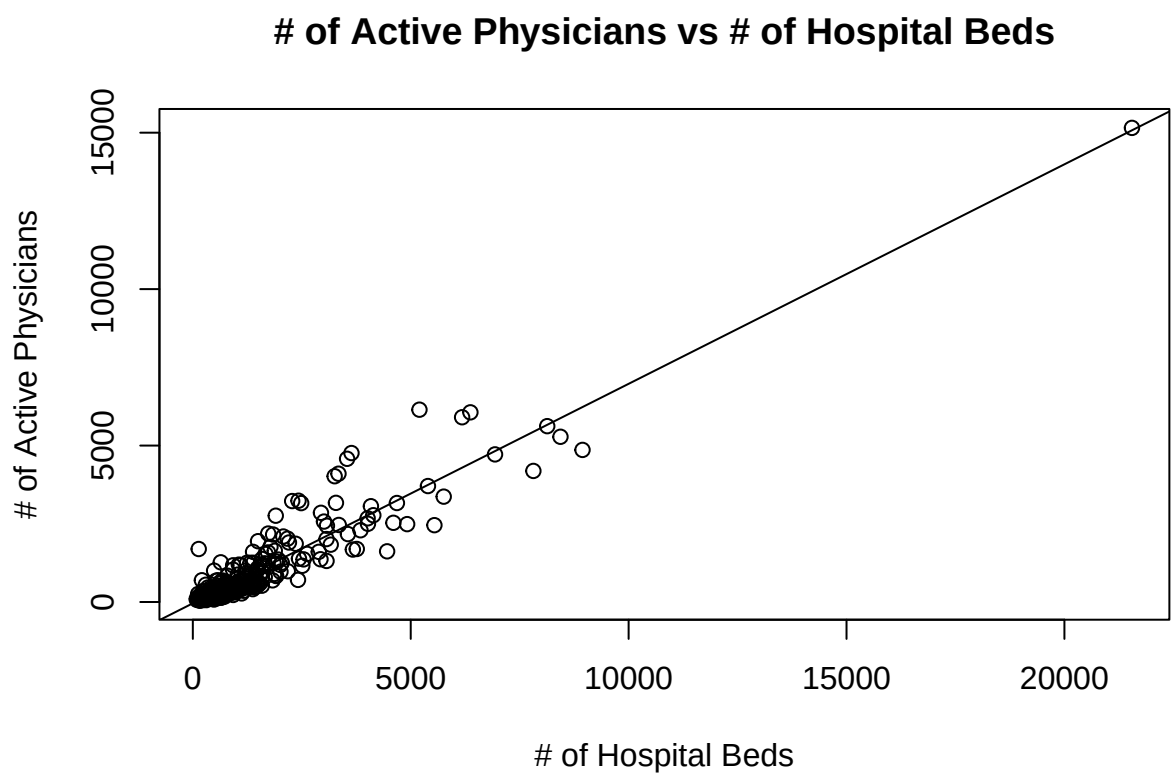
c)

```r
plot(train_set$X1, train_set$Y, main = "# of Active Physicians vs Total Population",
     xlab = "Total Population", ylab = "# of Active Physicians") + abline(fit_tp)
```



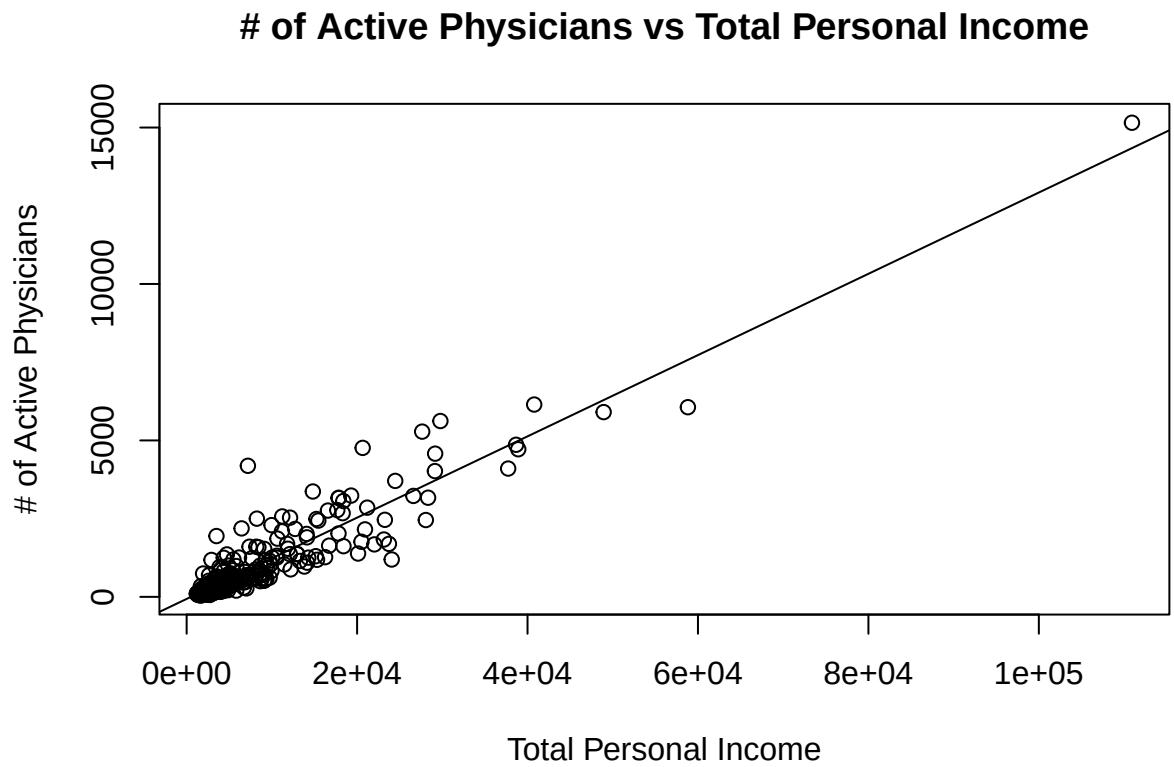

```
## integer(0)
```

```
plot(train_set$X2, train_set$Y, main = "# of Active Physicians vs # of Hospital Beds",
     xlab = "# of Hospital Beds", ylab = "# of Active Physicians") + abline(fit_hb)
```




```
## integer(0)
```

```r
plot(train_set$X3, train_set$Y, main = "# of Active Physicians vs Total Personal Income",
     xlab = "Total Personal Income", ylab = "# of Active Physicians") + abline(fit_pi)
```




```
## integer(0)
```

Because most of the data points lie close to the estimated regression line, each of the three linear regression appears to fit the data well. The R-squared values, 0.9029, 0.9073, 0.9241 (same order as the plots), also support this claim. We can observe a strong positive linear relationship between the number of active physicians and each of the three predictor variables.

d)

```r
n <- nrow(train_set)

MSE_tp <- sum(fit_tp$residuals^2)/(n-2)
MSE_hb <- sum(fit_hb$residuals^2)/(n-2)
MSE_pi <- sum(fit_pi$residuals^2)/(n-2)

MSE_tp
```

```
## [1] 299244.6
```

```r
MSE_hb
```

```
## [1] 238862
```

```r
MSE_pi
```

```
## [1] 242971
```

MSE (Total Population) = 377,001.2
MSE (# of Hospital Beds) = 359,910.6
MSE (Total Personal Income) = 294,367.6

Total.personal.income variable leads to the smallest variability around the fitted regression line. We will use this predictor variable to estimate Y because it gives the least MSE, given the training data. The variable that has the least MSE represent the data better than the others, so it is expected to provide us with a better prediction than the other variables with bigger MSE.

e)

```
summary(fit_pi)
```

```
##
## Call:
## lm(formula = Y ~ X3, data = train_set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1865.7  -176.7   -39.6    65.6  3321.4
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -65.584856  35.009926  -1.873    0.062 .
## X3            0.129883   0.002801  46.365   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 492.9 on 307 degrees of freedom
## Multiple R-squared:  0.875,  Adjusted R-squared:  0.8746
## F-statistic:  2150 on 1 and 307 DF,  p-value: < 2.2e-16
```

```
predicted_pi <- predict(fit_pi, test_set)

trained <- data.frame(obs = train_set$Y, pred = fit_pi$fitted.values)
defaultSummary(trained)
```

```
##        RMSE    Rsquared         MAE
## 491.3230426   0.8750341 287.5325187
```

```
tested <- data.frame(obs = test_set$Y, pred = predicted_pi)
defaultSummary(tested)
```

```
##        RMSE    Rsquared         MAE
## 721.7111345   0.9172166 350.8967035
```

(RMSE, Rsquared, MAE values are different in the knit file from those in RStudio. I used RStudio values as answers.) The linear regression model 'fit_pi' seems to be a very good model as R-squared is very high (0.0.9241). But, when this model is tested with the hold-out sample, R-squared decreases significantly to 0.7583. Considering the significant drop in R-squared, we should be worried about overfitting, and it seems further analysis is necessary. The p-value of X3 indicates that the predictor variable, Total.personal.income, is significant, and we can find a strong positive linear relationship between X3 and Y.

Problem 3

```
CDI_pi <- data.frame(cbind(Y,X3))

p3 <- function(dat0){
  n <- nrow(dat0)
```

```r
  out <- data.frame(matrix(0,nrow=1000,ncol=2))

  for(i in 1:1000){
    ind <-sample(1:n, 400)
    dat <- dat0[ind,]
    fit <- lm(Y ~ X3, data = dat)
    out[i,] <- c(fit$coefficients)
  }
  dimnames(out)[[2]] <- c("b0","b1")
  out
}

set.seed(1234)

ans <- p3(CDI_pi)
mean_beta <-apply(ans, 2, mean)
var_beta <- apply(ans, 2, var)

# Regression Model with Full Data
fit_full <- lm(Number.of.active.physicians ~ Total.personal.income, data = CDIdata)
b0_full <- fit_full$coefficients[1]
b1_full <- fit_full$coefficients[2]
```

From 1,000 regression models:

Mean of β_0 = -48.874
Mean of β_1 = 0.1318
Variance of β_0 = 73.6363
Variance of β_1 = 1.279e-06

β_0 of the full-data model: -48.3949
β_1 of the full-data model: 0.1317

As expected, the means of β_0 and β_1 calculated from 1,000 linear regressions are very close to the β_0 and β_1 of the regression model built with the full data. Because we took a big enough number (1,000) of random samples from the population, the mean of the sampled data is supposed to be very close to the population data.

```r
# Another way of getting the same results for means and variances of b0 and b1:
#
# N <- 1000
#
# set.seed(1234)
# betas <- replicate(N, {
#   dat <- CDIdata[sample(1:nrow(CDI_pi), 400),]
#   fit <- lm(Y ~ X3, data = dat)
#   b0 <- fit$coefficients[1]
#   b1 <- fit$coefficients[2]
#   beta <- c(b0, b1)
#   beta
# })
#
# mean_b0 <- mean(betas[1,])
# mean_b1 <- mean(betas[2,])
#
```

```r
# var_b0 <- sum((betas[1,]-mean_b0)^2)/(N-1)
# var_b1 <- sum((betas[2,]-mean_b1)^2)/(N-1)
```